TASK -2

Sensor-Based Simulation

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CODEALPHA

IoT LED ON/OFF using LDR Sensor in Tinkercad

- Circuit Components:
    1. Arduino UNO
    2. Breadboard
    3. LED
    4. Resistor (220 ohm)
    5. LDR Sensor
    6. Resistor (10 kohm)

- Arduino Code:

```
int ldr = A0;
int led = 13;
void setup() {
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}

void loop() {

   int value = analogRead(ldr);

   Serial.println(value);

   if(value < 500) {

       digitalWrite(led, HIGH);

    }
    else {
         digitalWrite(led, LOW);
    }

  delay(100);
}
```

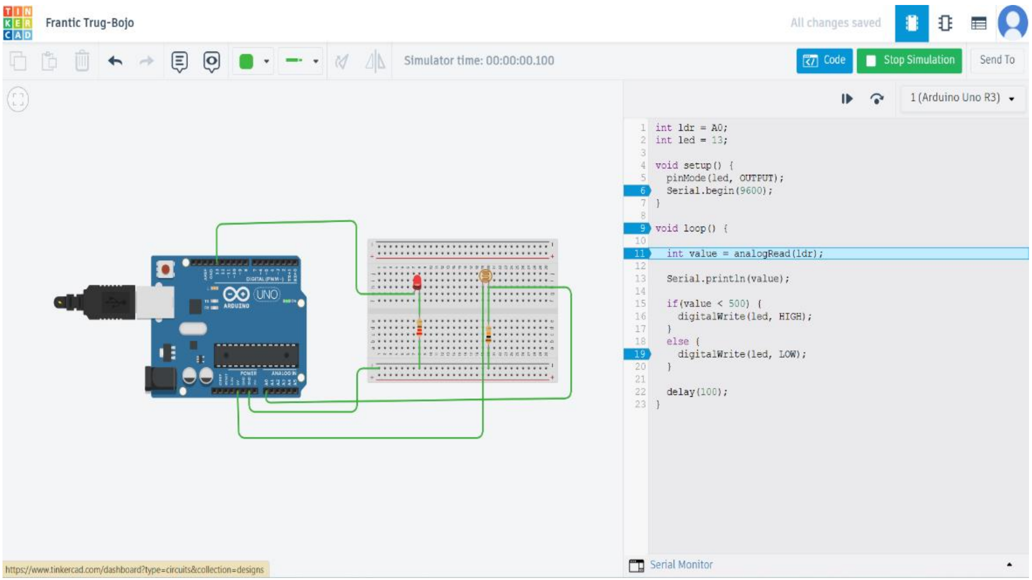
- ## Code Explanation:

  This IoT simulation uses an LDR sensor and an LED with Arduino UNO. The LDR detects light intensity and sends analog values to the Arduino through pin A0.

  If the light intensity becomes low (dark environment), the Arduino turns ON the LED connected to pin 13.
  If the light intensity is high (bright environment), the LED turns OFF.

  The project demonstrates automatic light control using sensors, which is a common IoT application in smart homes and smart lighting systems.

- ## Circuit Diagram



- Simulation Running:

Explanation in an image: